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Code JupyterLab Python [conda env:base] * Anaconda Toolbox

[1]:
```
%pip install tensorflow numpy matplotlibs
```

```
Requirement already satisfied: tensorflow in c:\users\anjali\anaconda3\lib\site-packages (2.21.0)
Requirement already satisfied: numpy in c:\users\anjali\anaconda3\lib\site-packages (1.26.4)
Requirement already satisfied: matplotlib in c:\users\anjali\anaconda3\lib\site-packages (3.9.2)
Requirement already satisfied: absl-py>=1.0.0 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (2.5.
1)
Requirement already satisfied: astunparse>=1.6.0 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (1.
6.3)
Requirement already satisfied: flatbuffers>=25.9.23 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow)
(25.12.19)
Requirement already satisfied: gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1 in c:\users\anjali\anaconda3\lib\site-packages (fr
om tensorflow) (0.7.0)
Requirement already satisfied: google_pasta>=0.1.1 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow)
(0.2.0)
Requirement already satisfied: libclang>=13.0.0 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (18.
1.1)
Requirement already satisfied: opt_einsum>=2.3.2 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (3.
4.0)
Requirement already satisfied: packaging in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (24.1)
Requirement already satisfied: protobuf<8.0.0,>=6.31.1 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflo
w) (7.36.2)
Requirement already satisfied: requests<3,>=2.21.0 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow)
(2.32.3)
Requirement already satisfied: setuptools in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (75.1.0)
Requirement already satisfied: six>=1.12.0 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (1.16.0)
Requirement already satisfied: termcolor>=1.1.0 in c:\users\anjali\anaconda3\lib\site-packages (from tensorflow) (3.
3.0)
```

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[2]:
```
import numpy as np
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
import matplotlib.pyplot as plt
```

[3]:
```
(x_train, y_train), (x_test, y_test) = keras.datasets.mnist.load_data()

# Normalize pixel values between 0 and 1
x_train = x_train.astype("float32") / 255.0
x_test = x_test.astype("float32") / 255.0

# Train only on normal images (digit 0)
x_train = x_train[y_train == 0]

# Keep only digits 0 and 1 in the test dataset
mask = np.isin(y_test, [0, 1])
x_test = x_test[mask]
y_test = y_test[mask]

print("Training data:", x_train.shape)
print("Testing data:", x_test.shape)
print("Test labels:", np.unique(y_test))
```

```
Training data: (5923, 28, 28)
Testing data: (2115, 28, 28)
Test labels: [0 1]
```

Test labels: [0 1]

[4]:
```
x_train = x_train.reshape(-1, 784)
x_test = x_test.reshape(-1, 784)

print("Training shape:", x_train.shape)
print("Testing shape:", x_test.shape)
```

```
Training shape: (5923, 784)
Testing shape: (2115, 784)
```

[5]:
```
input_layer = keras.Input(shape=(784,))

# Encoder
encoded = layers.Dense(128, activation="relu")(input_layer)
encoded = layers.Dense(32, activation="relu")(encoded)

# Decoder
decoded = layers.Dense(128, activation="relu")(encoded)
output_layer = layers.Dense(784, activation="sigmoid")(decoded)

autoencoder = keras.Model(input_layer, output_layer)

autoencoder.summary()
```

Model: "functional"

Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 784)	0
dense (Dense)	(None, 128)	100,480
dense_1 (Dense)	(None, 32)	4,128
dense_2 (Dense)	(None, 128)	4,224
dense_3 (Dense)	(None, 784)	101,136

Total params: 209,968 (820.19 KB)

Trainable params: 209,968 (820.19 KB)

Non-trainable params: 0 (0.00 B)

[6]:
```
autoencoder.compile(
    optimizer="adam",
    loss="mse",
    metrics=["mae"]
)
```

```
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
```


[7]:
```
history = autoencoder.fit(
    x_train,
    x_train,
    epochs=5,
    batch_size=128,
    validation_split=0.1
)
```

```
Epoch 1/5
42/42 ━━━━━━━━━━━━━━━━━━━━ 3s 17ms/step - loss: 0.1063 - mae: 0.2330 - val_loss: 0.0577 - val_mae: 0.1258
Epoch 2/5
42/42 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - loss: 0.0490 - mae: 0.1098 - val_loss: 0.0419 - val_mae: 0.0960
Epoch 3/5
42/42 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.0399 - mae: 0.0945 - val_loss: 0.0353 - val_mae: 0.0871
Epoch 4/5
42/42 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.0337 - mae: 0.0853 - val_loss: 0.0302 - val_mae: 0.0794
Epoch 5/5
42/42 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - loss: 0.0288 - mae: 0.0770 - val_loss: 0.0255 - val_mae: 0.0704
```

[8]:
```
reconstructed = autoencoder.predict(x_test)

error = np.mean((x_test - reconstructed) ** 2, axis=1)

threshold = np.percentile(error[y_test == 0], 95)

print("Anomaly threshold:", threshold)
```

```
67/67 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step
Anomaly threshold: 0.04662816654890775
```


[9]:
```
predictions = error > threshold

for i in range(10):
    if predictions[i]:
        print("Image", i, "- Anomaly")
    else:
        print("Image", i, "- Normal")
```

```
Image 0 - Anomaly
Image 1 - Normal
Image 2 - Anomaly
Image 3 - Normal
Image 4 - Normal
Image 5 - Anomaly
Image 6 - Normal
Image 7 - Normal
Image 8 - Anomaly
Image 9 - Anomaly
```

[10]:
```
plt.figure(figsize=(10, 4))

for i in range(5):
    # Original image
    plt.subplot(2, 5, i + 1)
    plt.imshow(x_test[i].reshape(28, 28), cmap="gray")
    plt.title("Original")
    plt.axis("off")
```

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```
    # Reconstructed image
    plt.subplot(2, 5, i + 6)
    plt.imshow(reconstructed[i].reshape(28, 28), cmap="gray")
    plt.title("Reconstructed")
    plt.axis("off")

plt.tight_layout()
plt.show()
```

